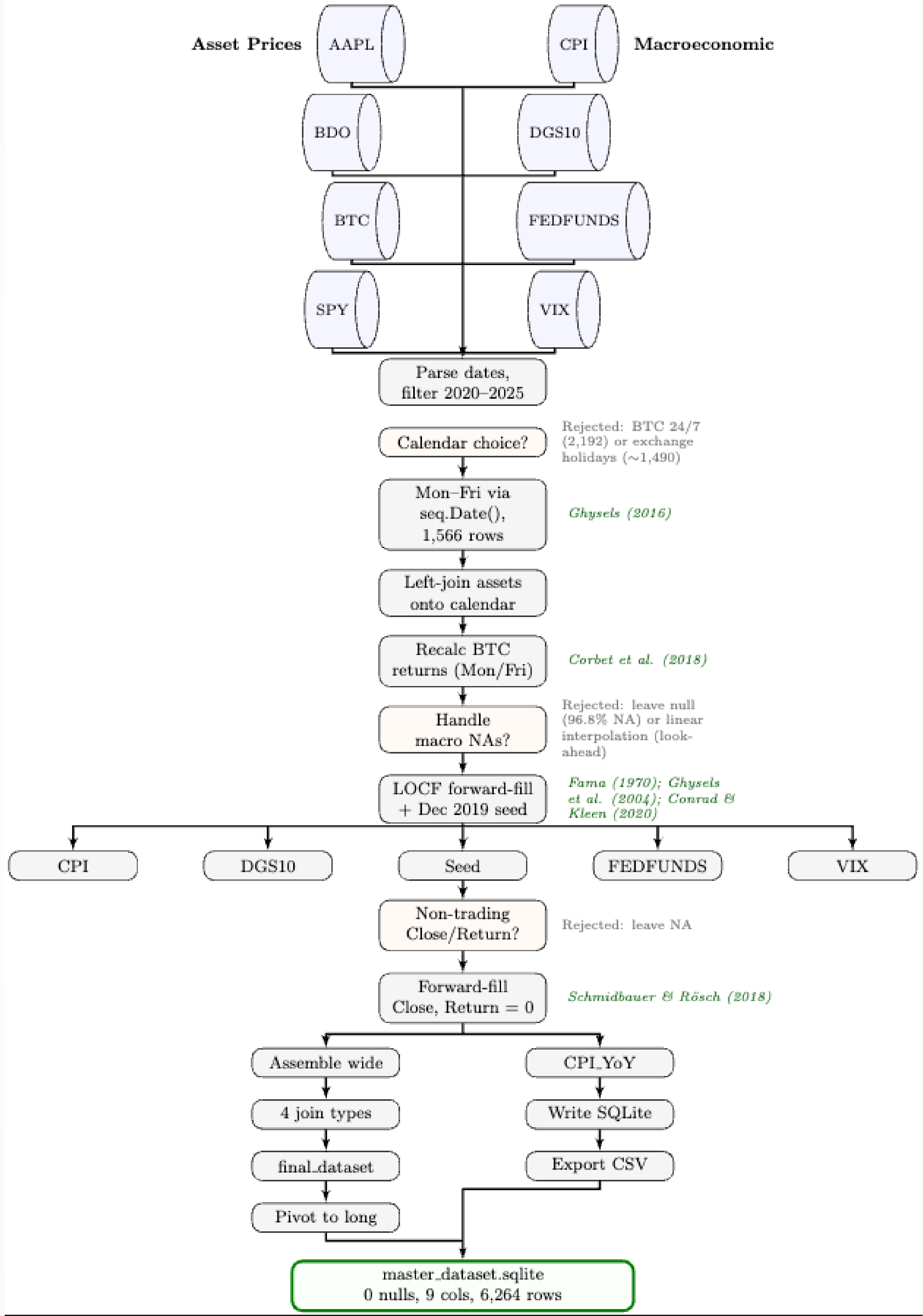
DATA CLEANING

- Unified three asset calendars (NYSE, PSE, 24/7) into one 1,566-day business calendar to remove exchange-specific holiday dependencies (Ghysels et al., 2016)
- LOCF-forward-filled monthly CPI and FEDFUNDS (step functions) to avoid interpolating non-existent prices (Fama, 1970)
- Recalculated BTC returns as business-day closes (Mon/Fri) since 24/7 formula used Sunday closes lost after weekend removal (Corbet et al., 2018)
- LOCF-forward-filled asset closes on exchange holidays with zero returns on non-trading days (Schmidbauer et al., 2018)
- Computed CPI YoY on raw monthly data pre-fill to avoid fill-induced inflation spikes
- All joins yielded identical 1,566-row outputs (zero unmatched); retained left_join to preserve asset calendar as base



Limitations

- Mon-Fri calendar overcounts tradable days by 10 to 15 days per year per exchange because exchange-specific holiday lists reduce reproducibility and introduce external dependencies.
- LOCF may lag the actual policy rate by up to two weeks during structural breaks such as emergency FOMC rate cuts because the last known observation remains the best estimate until the next scheduled release.
- Weekend cryptocurrency trading days discarded under the portfolio construction framework because weekend-only BTC price movements carry no economic weight for a portfolio constrained to equity market hours.

FINAL DATASET

Table 6: Observation Reduction Across Processing Stages

Dataset	Original	After Calendar Filter	After Forward-Fill	Final
AAPL	2512	1508	1566	—
BDO	2441	1468	1566	—
BTC	3652	1566	1566	—
SPY	2514	1508	1566	—
CPI	952	71	1566	—
DGS10	16 105	1500	1566	—
FEDFUNDS	863	72	1566	—
VIX	9216	1535	1566	—
Final Dataset	—	—	—	1566

SQL Section Q1

Which asset delivered the highest return per unit of risk over the 2020 to 2025 sample period?

- Computed a Sharpe-like ratio by dividing each asset's mean daily return by its standard deviation.
- Sorted assets descending: BDO (0.0524), BTC (0.0520), AAPL (0.0392), SPY (0.0353).
- Constructed the metric via `group_by(Asset) |> summarise(Mean_Daily_Return, SD_Daily_Return) |> mutate(Risk_Adjusted_Return)`.
- Extended the query with `arrange(desc(Risk_Adjusted_Return))` to produce the final ranking.
- Added a benchmark reference line at SPY's ratio (0.0353) to contextualize relative performance.

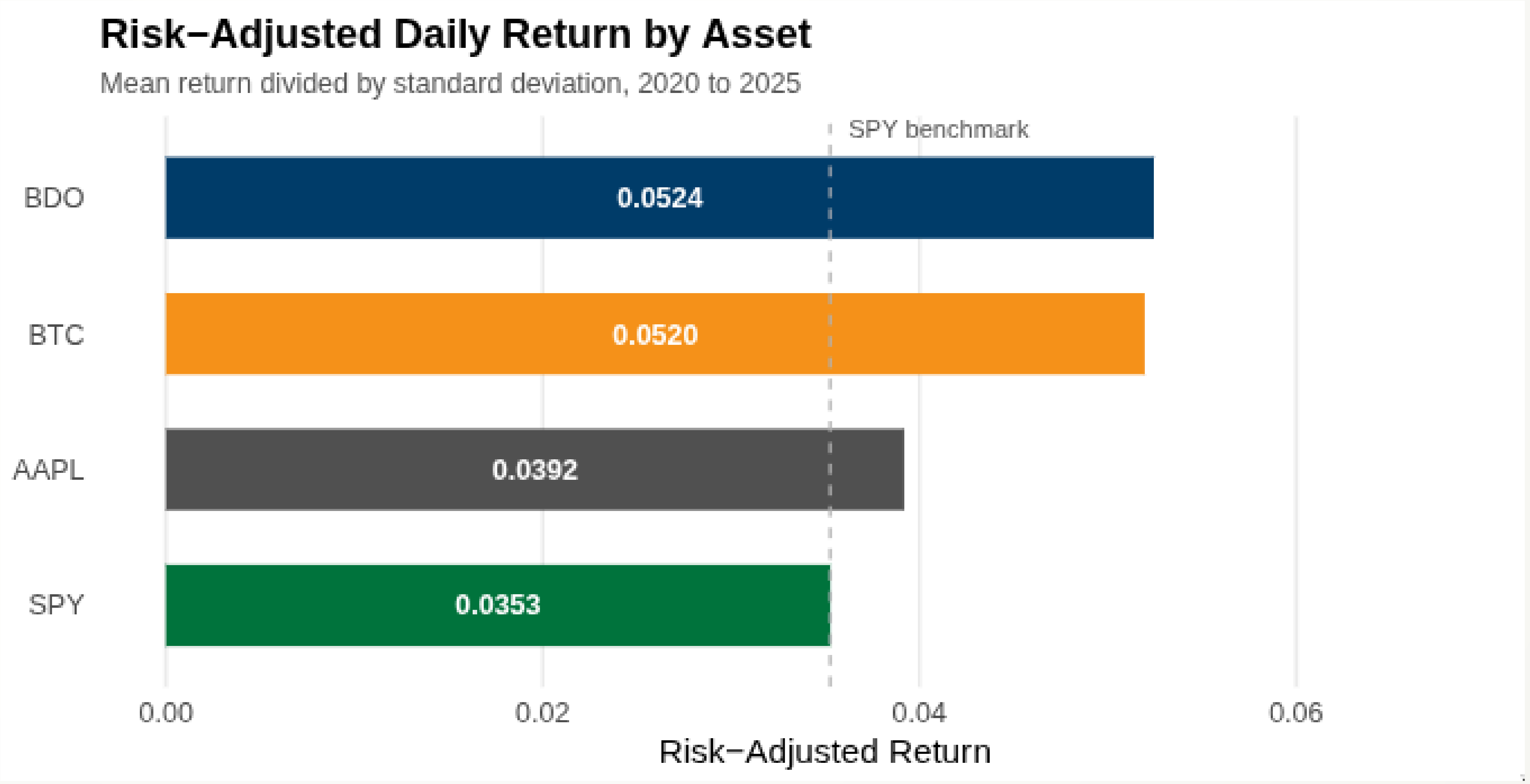
Table 7: Q1 Data Output

Asset	Mean_Daily_Return	SD_Daily_Return	Risk_Adjusted_Return
BDO	0.0008	0.0154	0.0524
BTC	0.0017	0.0319	0.0520
AAPL	0.0006	0.0144	0.0392
SPY	0.0003	0.0095	0.0353

SQL Section Q1

Figure 1 plots the risk-adjusted return as horizontal bars, ranked top to bottom by efficiency.

- A dashed vertical line at SPY's benchmark level divides beaters from laggards.
- BDO led the hierarchy at 0.0524, built on a 0.08% mean daily return against 1.54% volatility.
- BTC followed at 0.0520, achieving near-parity through higher return magnitude (0.17%) at 3.19% volatility.
- AAPL and SPY ranked below the Philippine asset, with AAPL at 0.0392 and SPY at 0.0353.
- BDO delivered the highest return per unit of risk, driven by its low volatility rather than high absolute returns.



SQL Section Q2

How did each asset perform during the COVID-19 market crash of February to March 2020?

- Filtered the dataset to observations between 1 February and 31 March 2020.
- Computed cumulative returns within each asset group using `cumprod(1 + Daily_Return) - 1` ordered by date.
- Summarised start and end dates, total cumulative return, and observation count per asset.
- BDO posted a positive cumulative return of +41.67% across 60 trading days, gaining during the crisis.
- BTC declined 31.14% across 60 days, the largest cumulative loss.
- SPY lost 4.46% and AAPL lost 0.43%, both over 60 trading days.
- BDO's positive COVID return reflected the delayed and muted impact of the global selloff on Philippine equities rather than genuine crisis alpha.

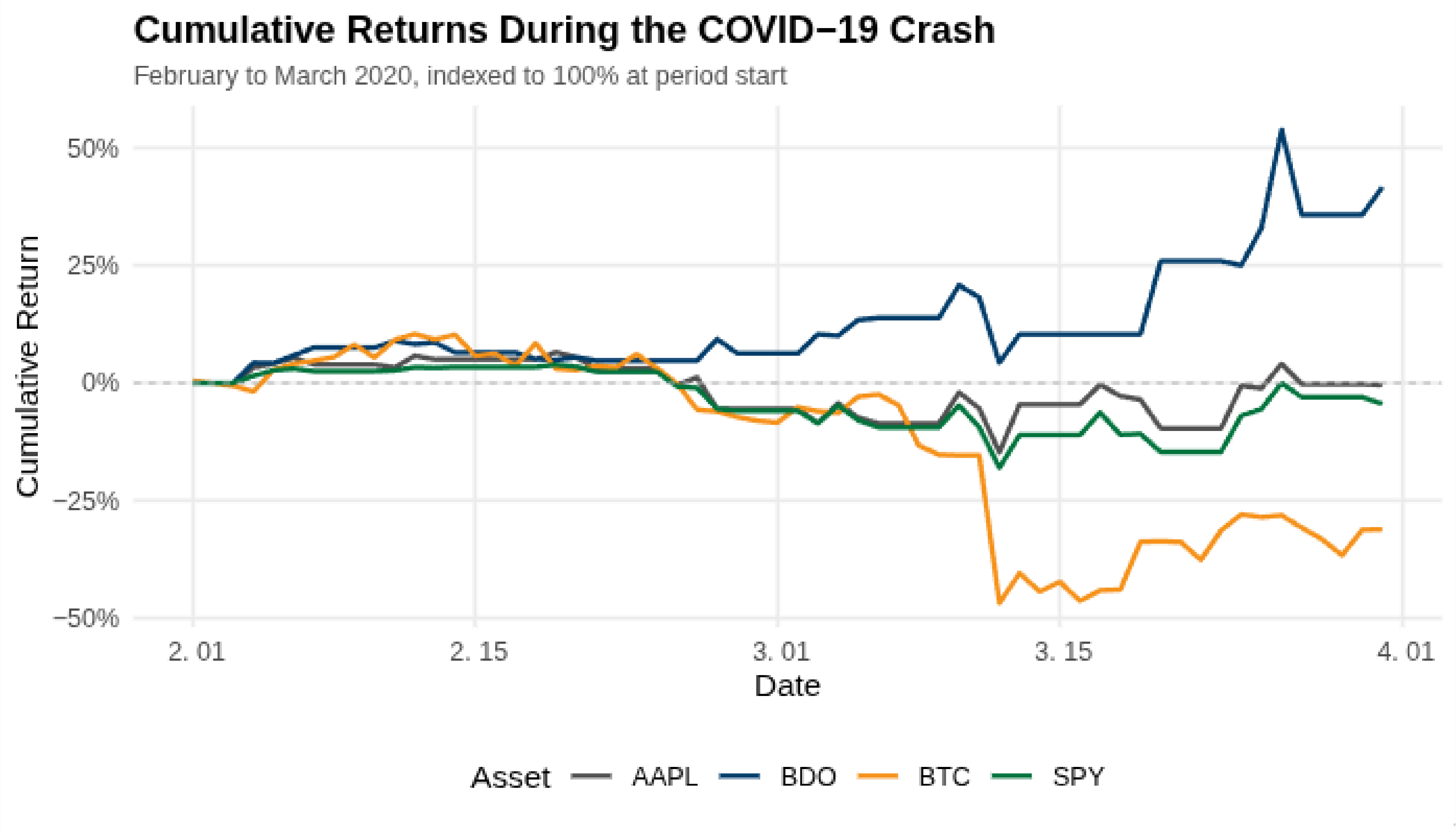
Table 8: Q2 Summary Data Output

Asset	Start_Date	End_Date	Total_Cumulative_Return	Observations
AAPL	2020-02-01	2020-03-31	-0.0043	60
BDO	2020-02-01	2020-03-31	0.4167	60
BTC	2020-02-01	2020-03-31	-0.3114	60
SPY	2020-02-01	2020-03-31	-0.0446	60

SQL Section Q2

Figure 2 traces cumulative returns as line paths from February through March 2020.

- SPY and AAPL declined modestly, reflecting the broad US equity selloff compressed into a narrow window.
- BTC exhibited sharp intra-period reversals with violent single-day bounces followed by deeper declines.
- BDO showed a pronounced upward trajectory that contrasted sharply with the US equity collapse.
- Confirmed that the proposition of cryptocurrency as a crisis hedge is rejected — BTC recorded the largest cumulative loss at 31.14%.
- Answered the research question: crisis impact was highly asset-specific, with BDO gaining as Philippine markets decoupled from global turmoil.



SQL Section Q3

What is the cross-asset correlation of daily returns, and which assets offer the strongest diversification potential?

- Transformed daily returns into wide format via `pivot_wider(names_from = Asset, values_from = Daily_Return)`.
- Computed the Pearson correlation matrix using `cor(use = "pairwise.complete.obs")`.
- Reshaped the matrix to long format via `pivot_longer` for visual mapping.
- Found the AAPL-SPY pair produced the highest coefficient at 0.787, a mechanical consequence of Apple's weighting in the S&P 500.
- BDO formed a distinct diversification tier with average cross-correlation of 0.090.

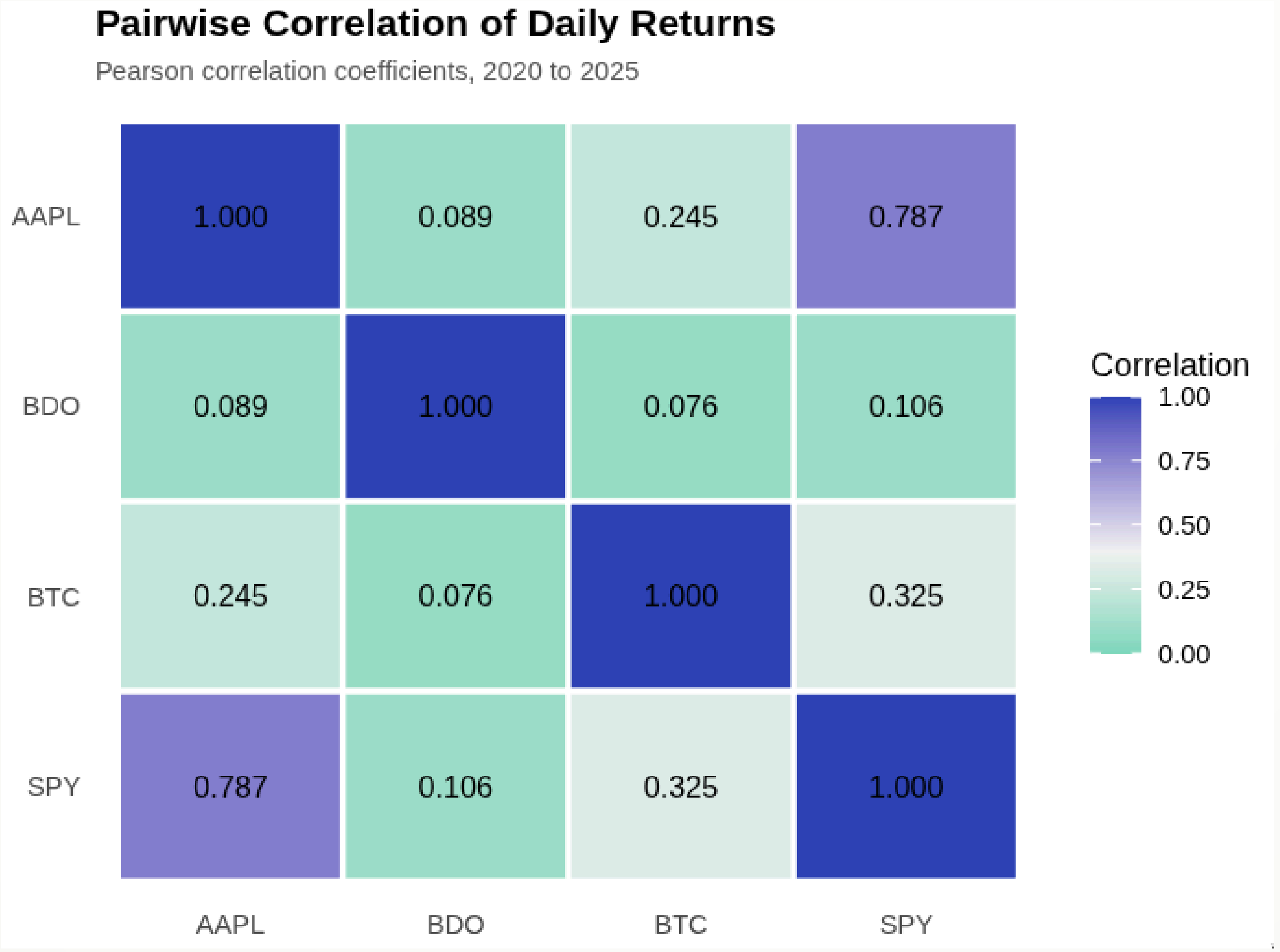
Table 9: Q3 Correlation Matrix Output

	AAPL	BDO	BTC	SPY
AAPL	1.0000	0.0894	0.2451	0.7871
BDO	0.0894	1.0000	0.0760	0.1056
BTC	0.2451	0.0760	1.0000	0.3250
SPY	0.7871	0.1056	0.3250	1.0000

SQL Section Q3

Figure 3 displays the pairwise correlation matrix as a colour-coded heatmap with numeric labels.

- Dark blue squares mark the strongest positive correlations (AAPL-SPY at 0.787), light green squares mark the near-zero pairs.
- BDO's average cross-correlation of 0.090 means it retains more than 91% of its stand-alone variance as diversifying power.
- The BDO-BTC pair at 0.076 was the lowest in the matrix, offering the strongest portfolio diversification.
- BDO provides the greatest diversification benefit, but unconditional correlations overstate the protection available during systemic crises.



SQL Section Q4

How do the selected assets rank by total return from 2020 to 2025?

- Extracted start and end closing prices per asset from the filtered daily dataset.
- Computed total return as `(End_Close / Start_Close) - 1` and annualised return via the 365.25-day formula.
- Ranked by `Total_Return` descending to produce the performance hierarchy.
- BTC delivered 1,115.4% total return (51.6% annualised), far exceeding all other assets.
- BDO returned 5.7% total return (0.9% annualised), the lowest in the sample — essentially zero real capital appreciation across six years.
- All assets have 2,192 observations after the forward-fill procedure.

Table 10: Q4 Data Output

Asset	Start_Date	End_Date	Start_Close	End_Close	Total_Return	Annualized_Return	Mean_Daily_Return	SD_Daily_Return	Obs
BTC	2020-01-01	2025-12-31	7200.17	87508.83	11.1537	0.5164	0.0017	0.0319	2192
AAPL	2020-01-01	2025-12-31	72.33	271.36	2.7517	0.2466	0.0006	0.0144	2192
SPY	2020-01-01	2025-12-31	296.13	678.32	1.2906	0.1482	0.0003	0.0095	2192
BDO	2020-01-01	2025-12-31	127.35	134.60	0.0569	0.0093	0.0008	0.0154	2192

SQL Section Q8

Does elevated trading volume systematically coincide with higher daily price volatility?

- Filtered to observations with non-missing VIX values and partitioned into three regimes: VIX < 20, 20-30, and VIX > 30.
- Grouped by Asset and VIX_Regime to compute conditional mean return, standard deviation, and observation count.
- During high-fear periods, BDO was the only asset with a positive mean daily return (0.14%), while AAPL posted -0.34%, SPY -0.24%, and BTC -0.16%.
- BTC's negative high-fear mean contradicted the narrative that cryptocurrency hedges volatility.
- The high-fear regime contained 206 observations per asset, approximately 9.4% of the total sample.

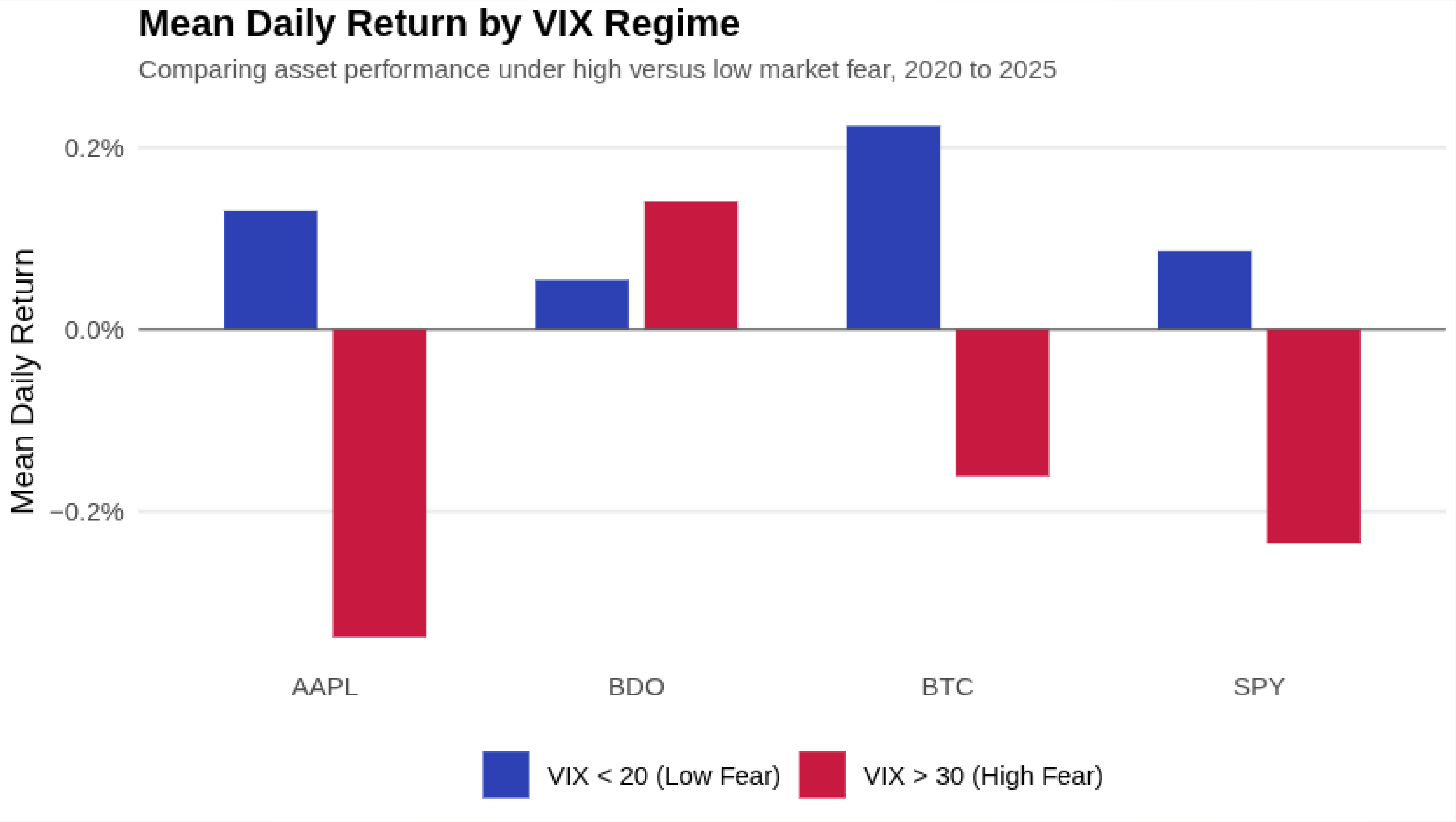
Table 14: Q8 Data Output

Asset	VIX_Regime	Mean_Daily_Return	SD_Daily_Return	Obs
AAPL	High Volatility (VIX > 30)	-0.0034	0.0283	206
AAPL	Low Volatility (VIX < 20)	0.0013	0.0096	1221
AAPL	Moderate (20 <= VIX <= 30)	0.0004	0.0151	765
BDO	High Volatility (VIX > 30)	0.0014	0.0263	206
BDO	Low Volatility (VIX < 20)	0.0005	0.0131	1221
BDO	Moderate (20 <= VIX <= 30)	0.0011	0.0149	765
BTC	High Volatility (VIX > 30)	-0.0016	0.0495	206
BTC	Low Volatility (VIX < 20)	0.0022	0.0261	1221
BTC	Moderate (20 <= VIX <= 30)	0.0016	0.0342	765
SPY	High Volatility (VIX > 30)	-0.0024	0.0220	206
SPY	Low Volatility (VIX < 20)	0.0009	0.0052	1221
SPY	Moderate (20 <= VIX <= 30)	0.0002	0.0092	765

SQL Section Q8

Figure 8 plots mean daily return by VIX regime as a grouped bar chart, blue for low fear and red for high fear.

- BDO's bar was the only positive red bar — every other asset's high-fear bar extended below the zero line.
- AAPL's low-fear mean (0.13%) was substantially higher than BDO's (0.05%), establishing a clear calm-market hierarchy.
- The COVID-19 crash from Q2 drove the high-fear regime, with BDO's positive return reflecting its decoupling from global risk-on dynamics.
- Answered the research question: VIX regime determines cross-asset return rankings — during high fear, BDO uniquely delivered positive returns, likely reflecting domestic monetary insulation rather than genuine crisis alpha.



SQL Section Q9

How sensitive is each asset's daily return to changes in the 10-year Treasury yield?

- Computed daily changes in the 10-year yield via `DGS10_Change = DGS10 - lag(DGS10)` per asset.
- Assigned each day to Yields Rising, Yields Falling, or No Change based on the sign of the change.
- Grouped by Asset and Yield_Direction to produce conditional mean returns.
- During rising-yield periods, all four assets posted positive returns — BTC (0.20%), BDO (0.13%), AAPL (0.11%), SPY (0.10%).
- During falling-yield periods, BDO posted 0.12% and BTC 0.17%, while SPY was near-zero at 0.01%.

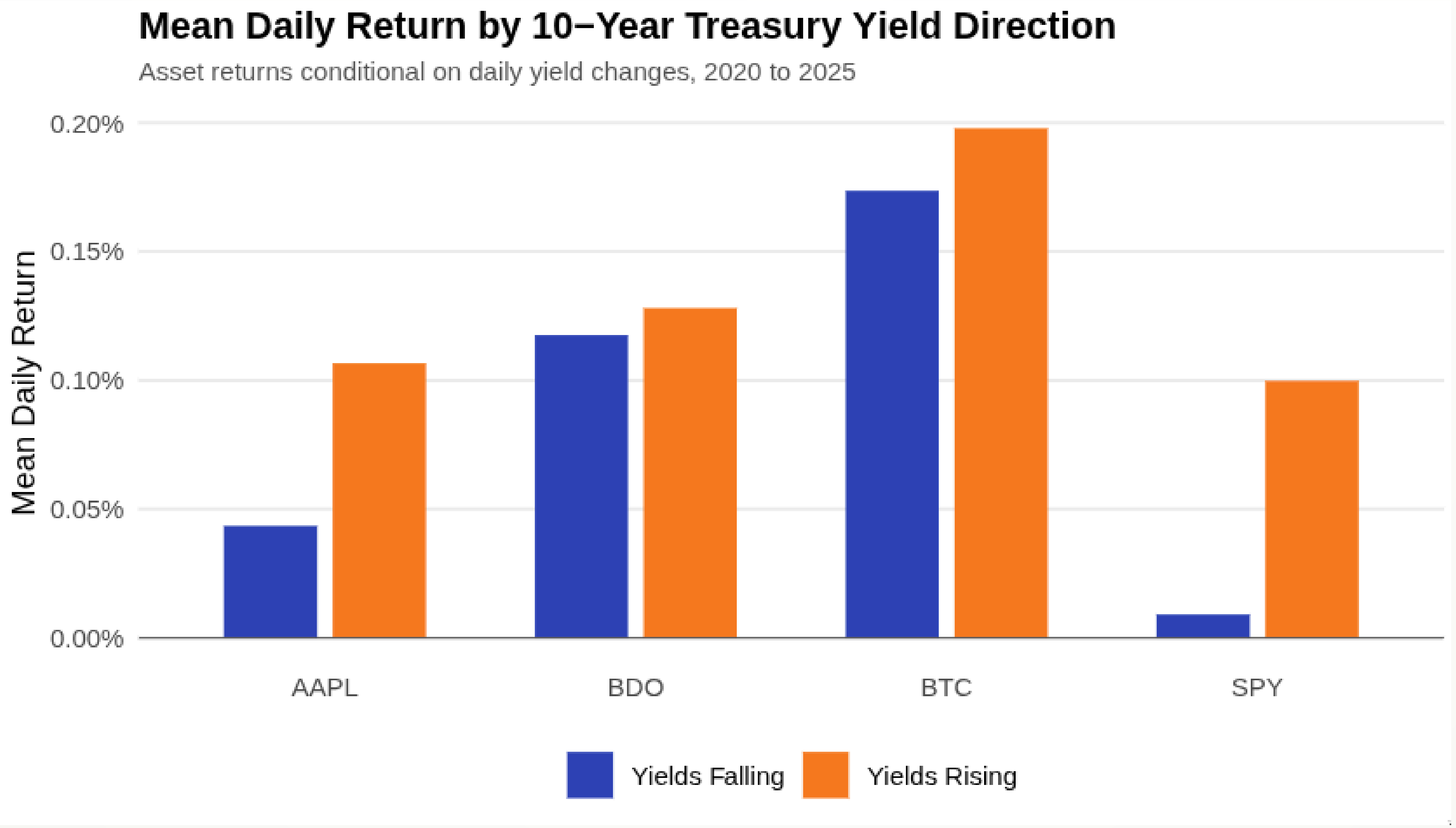
Table 15: Q9 Data Output

Asset	Yield_Direction	Mean_Daily_Return	SD_Daily_Return	Obs
AAPL	No Change	0.0002	0.0068	795
AAPL	Yields Falling	0.0004	0.0170	684
AAPL	Yields Rising	0.0011	0.0176	712
BDO	No Change	0.0001	0.0058	795
BDO	Yields Falling	0.0012	0.0203	684
BDO	Yields Rising	0.0013	0.0172	712
BTC	No Change	0.0013	0.0248	795
BTC	Yields Falling	0.0017	0.0343	684
BTC	Yields Rising	0.0020	0.0363	712
SPY	No Change	-0.0000	0.0035	795
SPY	Yields Falling	0.0001	0.0109	684
SPY	Yields Rising	0.0010	0.0123	712

SQL Section Q9

Figure 9 plots mean daily return conditional on yield direction, orange for rising yields and blue for falling yields.

- Every asset shows a positive orange bar, contradicting the textbook DCF intuition that higher rates mechanically lower prices.
- The growth-signalling channel dominated the discount-rate channel: rising yields reflected improving growth expectations, not restrictive monetary policy.
- BDO showed positive returns in both regimes (0.13% rising, 0.12% falling), the most consistent performance across yield environments.
- Answered the research question: rate sensitivity varies by asset, but the dominant transmission channel is growth expectations.



SQL Section Q10

Which assets served as the most effective inflation hedges during the high-inflation period of 2021 to 2023?

- Filtered the dataset to the window where CPI year-on-year exceeded 5% (June 2021 to February 2023).
- Computed cumulative returns per asset using `cumprod(1 + Daily_Return) - 1` within the filtered window.
- Summarised start and end dates, CPI boundaries, total cumulative return, and mean daily return per asset.
- AAPL generated the highest cumulative return at 27.8%, followed by BDO at 24.8%.
- SPY gained 11.3%, while BTC collapsed by 38.0% — the asset marketed as digital gold delivered the worst inflation-hedging performance.
- All four assets share 638 high-inflation observations.

Table 16: Q10 Data Output

Asset	Start_Date	End_Date	Start_CPI_YoY	End_CPI_YoY	Total_Cumulative_Return	Mean_Daily_Return	Obs
AAPL	2021-06-01	2023-02-28	0.0530	0.0596	0.2778	0.0005	638
BDO	2021-06-01	2023-02-28	0.0530	0.0596	0.2481	0.0004	638
SPY	2021-06-01	2023-02-28	0.0530	0.0596	0.1130	0.0002	638
BTC	2021-06-01	2023-02-28	0.0530	0.0596	-0.3800	-0.0002	638

SQL Section Q10

Figure 10 plots cumulative return during the high-inflation window as horizontal bars, ranked best to worst.

- AAPL's bar leads at +27.8%, BDO follows at +24.8%, SPY at +11.3%, and BTC extends deep into negative territory at -38.0%.
- AAPL's pricing power allowed it to pass input cost increases to consumers without triggering demand destruction.
- BDO's performance reflected the endowment effect: Philippine banks widen net interest margins when floating-rate loans reprice upward while deposit rates adjust slowly.
- BTC's 38.0% loss was the most informative negative data point: no cash flows, no contractual repricing mechanism, and no fundamental valuation floor meant the asset collapsed when the Fed withdrew liquidity.
- Answered the research question: AAPL and BDO provided the most effective inflation hedges, while the asset with the highest stand-alone total return in Q4 became the worst performer during the inflation window.